

Pharmacotherapy 2: Ultimate Question Bank

Batch 2: Lecture 1 & 2 (NSAIDs, Acetaminophen & Gout)

50 High-Yield Exam Questions with Clinical Explanations

1. Which chemical mediators are released upon tissue damage to stimulate pain receptors?

A) Prostaglandins and Leukotrienes

B) Histamine and Bradykinin

C) Acetylcholine and Dopamine

D) Serotonin and GABA



Hint: As the doctor explained with the 'nail' example, tissue injury causes immediate release of Histamine and Bradykinin which trigger pain. Prostaglandins only 'sensitize' the receptors.

2. What is the primary role of Prostaglandins (PGs) in the context of pain perception?

A) They directly cause severe pain

B) They sensitize nerve endings to Histamine and Bradykinin

C) They block pain signals at the spinal cord

D) They reduce inflammation immediately



Hint: Prostaglandins do not cause pain directly; they lower the pain threshold (Sensitization) so you feel the pain from bradykinin/histamine more intensely.

3. NSAIDs reduce pain primarily by inhibiting the synthesis of which substance?

A) Histamine

B) Bradykinin

C) Prostaglandins

D) Leukotrienes



Hint: By inhibiting COX, NSAIDs prevent the synthesis of Prostaglandins, removing the sensitization effect on nerve endings.

4. The Arachidonic Acid pathway is initiated by the release of Arachidonic Acid from cell membrane phospholipids via which enzyme?

- A) Cyclooxygenase
- B) Lipoxygenase
- C) Phospholipase A2**
- D) Thromboxane synthase

 **Hint:** Phospholipase A2 acts on membrane phospholipids to release Arachidonic acid, the raw material for both COX and LOX pathways.

5. Which enzymatic pathway converts Arachidonic Acid into Leukotrienes?

- A) COX Pathway
- B) LOX Pathway**
- C) P450 Pathway
- D) Glucuronidation Pathway

 **Hint:** The Lipoxygenase (LOX) pathway produces Leukotrienes, which are potent bronchoconstrictors.

6. Why can NSAIDs precipitate bronchospasm in asthmatic patients?

- A) They inhibit Leukotriene synthesis
- B) They shunt Arachidonic Acid to the LOX pathway, increasing Leukotrienes**
- C) They increase Histamine release
- D) They block Beta-2 receptors

 **Hint:** This is the 'Asthma Trap'. Blocking COX forces all Arachidonic Acid into the LOX pathway, massively producing bronchoconstricting Leukotrienes.

7. COX-1 is considered a 'housekeeping' enzyme because it is constitutively present in:

A) Inflamed tissues only

B) GIT, Platelets, and Kidneys

C) Brain and Liver

D) Lungs and Heart



Hint: Dr. emphasized COX-1 protects the stomach (increases mucus, decreases acid) and maintains normal platelet and renal function.

8. Inhibition of COX-1 leads to which adverse effects?

A) Reduced Pain and Fever

B) Gastric Ulcers, Bleeding, and Renal issues

C) Bronchoconstriction

D) Sedation



Hint: Since COX-1 is protective, blocking it (with non-selective NSAIDs) removes gastric protection, leading to ulcers and bleeding.

9. Which of the following is a non-selective NSAID that reversibly inhibits both COX-1 and COX-2?

A) Aspirin

B) Celecoxib

C) Ibuprofen

D) Paracetamol



Hint: Ibuprofen is a classic reversible, non-selective NSAID. Aspirin is irreversible, and Celecoxib is COX-2 selective.

10. What makes Aspirin unique among NSAIDs regarding its mechanism of action?

- A) It selectively inhibits COX-2
- B) It irreversibly inhibits COX-1 and COX-2 via acetylation**
- C) It does not affect platelets
- D) It acts centrally only

 **Hint:** Aspirin is 'unique' because it covalently acetylates the COX enzyme, destroying it permanently until a new enzyme is synthesized.

11. How long does the antiplatelet effect of Aspirin last?

- A) 4-6 hours
- B) 24 hours
- C) 7-10 days**
- D) Until renal excretion

 **Hint:** Because platelets lack nuclei, they cannot synthesize new COX. The effect lasts for the entire lifetime of the platelet (7-10 days).

12. Low dose Aspirin (75-81 mg/day) is primarily used for:

- A) Anti-inflammatory effect
- B) Analgesia
- C) Antiplatelet prophylaxis**
- D) Antipyresis

 **Hint:** Low doses selectively inhibit Thromboxane A₂ (TXA₂) in platelets without significantly affecting systemic PGI₂, acting as a prophylactic against thrombosis.

13. High dose Aspirin is typically indicated for:

- A) Stroke prevention
- B) Rheumatoid Arthritis (Anti-inflammatory)**
- C) Mild headache
- D) Fever reduction



Hint: Anti-inflammatory effects of Aspirin are only achieved at much higher doses, useful for conditions like Rheumatoid Arthritis.

14. Selective COX-2 inhibitors like Celecoxib have an advantage over non-selective NSAIDs regarding:

- A) Cardiovascular safety
- B) GI toxicity (Ulcers/Bleeding)**
- C) Renal safety
- D) Cost



Hint: By leaving COX-1 alone, Celecoxib maintains gastric mucosal protection, significantly reducing the risk of ulcers.

15. What is a major disadvantage of selective COX-2 inhibitors?

- A) Increased risk of GI ulcers
- B) Potential increased risk of Cardiovascular events (Thrombosis)**
- C) Lack of analgesic effect
- D) Severe sedation



Hint: They inhibit the vasodilatory PGI₂ (COX-2 mediated) but leave the aggregatory TXA₂ (COX-1 mediated) unopposed, increasing MI and stroke risks.

16. Paracetamol differs from true NSAIDs because it has:

- A) Strong anti-inflammatory activity
- B) Weak/No Anti-inflammatory effect**
- C) Irreversible COX inhibition
- D) Significant GI toxicity

 **Hint:** Paracetamol works mainly in the CNS. In the periphery, it is inactivated by peroxides in inflamed tissues, so it has negligible anti-inflammatory action.

17. Why is Paracetamol preferred over Aspirin for fever in children?

- A) It is more effective
- B) It does not cause Reye's Syndrome**
- C) It has stronger anti-inflammatory action
- D) It prevents seizures

 **Hint:** Aspirin is absolutely contraindicated in children with viral infections (e.g., flu, chickenpox) due to the fatal risk of Reye's syndrome.

18. Diflunisal is an NSAID that lacks which therapeutic effect?

- A) Analgesic
- B) Anti-inflammatory
- C) Antipyretic**
- D) Antiplatelet

 **Hint:** Diflunisal cannot cross the Blood-Brain Barrier (BBB), so it cannot reach the hypothalamus to exert an antipyretic effect.

19. Why does Diflunisal lack antipyretic effect?

A) It does not inhibit COX enzymes

B) It does not cross the Blood-Brain Barrier (BBB)

C) It is rapidly metabolized

D) It acts peripherally only



Hint: As mentioned by the doctor, the antipyretic action requires the drug to reach the thermoregulatory center in the brain, which Diflunisal fails to do.

20. Do NSAIDs arrest disease progression or induce remission in Rheumatoid Arthritis?

A) Yes, they cure RA

B) No, they only manage symptoms

C) Only high doses do

D) Only when combined with steroids



Hint: NSAIDs are purely symptomatic treatments. They reduce pain and inflammation but do not stop the underlying joint destruction in RA.

21. NSAIDs are generally less effective for which type of pain unless combined with opioids?

A) Musculoskeletal pain

B) Visceral pain

C) Headache

D) Dysmenorrhea



Hint: NSAIDs are excellent for somatic/musculoskeletal pain but weak against deep visceral pain (like organ pain), where opioids are often needed.

22. How do NSAIDs reduce fever?

A) By causing hypothermia

B) By resetting the hypothalamic set point via PGE2 inhibition

C) By increasing sweating only

D) By blocking histamine



Hint: Pyrogens increase PGE2 in the hypothalamus, which raises the temperature 'set-point'. NSAIDs block this PGE2 synthesis, resetting the thermostat.

23. Do NSAIDs cause hypothermia in normal individuals with normal body temperature?

A) Yes, significantly

B) No, they only lower elevated temperature

C) Only in elderly

D) Only with high doses



Hint: NSAIDs have ZERO effect on normal body temperature. They only work when the hypothalamic set-point is abnormally elevated.

24. GI toxicity from NSAIDs is caused by inhibition of COX-1 leading to decreased secretion of:

A) Hydrochloric acid

B) Protective Prostaglandins (PGE2, PGI2), Mucus, and Bicarbonate

C) Pepsinogen

D) Intrinsic factor



Hint: COX-1 produces the prostaglandins that stimulate mucus/bicarbonate production and inhibit acid secretion. Blocking it removes this vital protective shield.

25. Which drug is a PGE1 analog used to prevent NSAID-induced gastric ulcers?

- A) Omeprazole
- B) Misoprostol**
- C) Ranitidine
- D) Sucralfate



Hint: Misoprostol directly replaces the missing protective prostaglandins, effectively preventing NSAID-induced ulcers, though it causes diarrhea and uterine contractions.

26. Proton Pump Inhibitors (PPIs) like Omeprazole protect against NSAID gastropathy by:

- A) Increasing mucus secretion
- B) Decreasing gastric acid secretion**
- C) Neutralizing stomach acid
- D) Coating the ulcer



Hint: PPIs are often the preferred agents to co-administer with NSAIDs because they profoundly suppress acid secretion, allowing the mucosa to heal.

27. Physiological Prostaglandins (PGE2, PGI2) maintain Renal Blood Flow (RBF) by causing vasodilation of:

- A) Efferent Arterioles
- B) Afferent Arterioles**
- C) Glomerular capillaries
- D) Peritubular capillaries



Hint: Prostaglandins dilate the Afferent arteriole (bringing blood into the kidney). NSAIDs constrict it, heavily reducing blood flow to the kidney.

28. NSAID-induced Acute Kidney Injury (AKI) is most likely to occur in patients with:

A) Hypertension

B) Dehydration/Hypovolemia

C) Diabetes

D) Obesity

 **Hint:** In dehydrated patients, the kidney relies entirely on prostaglandins to keep the afferent arteriole open. Giving an NSAID in this state causes rapid renal failure.

29. Chronic NSAID use can lead to sodium and water retention resulting in:

A) Hypotension

B) Edema and Hypertension

C) Polyuria

D) Hyperkalemia

 **Hint:** By inhibiting renal PGs, NSAIDs promote sodium and water reabsorption, which can exacerbate hypertension and cause peripheral edema.

30. NSAIDs should be avoided in patients with Heart Failure primarily due to risk of:

A) Arrhythmias

B) Fluid retention and worsening failure

C) Myocardial infarction

D) Valve dysfunction

 **Hint:** The sodium and water retention caused by NSAIDs increases blood volume (preload), which the failing heart cannot pump effectively, worsening the heart failure.

31. High doses of Aspirin initially cause Central Respiratory Stimulation leading to:

A) Metabolic Acidosis

B) Respiratory Alkalosis

C) Respiratory Acidosis

D) Metabolic Alkalosis



Hint: Aspirin directly stimulates the respiratory center causing hyperventilation (washing out CO₂), which initially results in Respiratory Alkalosis.

32. Severe Salicylate toxicity (>50 mg/dL) is characterized by:

A) Respiratory Alkalosis only

B) Metabolic Acidosis, Hyperthermia, Coma

C) Hypoglycemia

D) Bradycardia



Hint: At toxic levels, Aspirin uncouples oxidative phosphorylation, leading to massive heat production (hyperthermia), accumulation of acids (metabolic acidosis), and coma.

33. Reye's Syndrome is associated with the use of Salicylates in children with viral infections like:

A) Measles and Mumps

B) Chickenpox or Influenza

C) Rubella

D) Hepatitis A



Hint: The classical link for Reye's syndrome is giving Aspirin to a child recovering from Varicella (chickenpox) or Influenza.

34. Reye's Syndrome primarily affects which two organs?

- A) Heart and Lungs
- B) Liver and Brain**
- C) Kidneys and Pancreas
- D) Spleen and Bone Marrow

 **Hint:** It is characterized by fulminant hepatitis (liver damage) and cerebral edema (encephalopathy in the brain).

35. Mild Salicylate toxicity (>30 mg/dL) typically presents with:

- A) Coma
- B) Tinnitus and Hyperventilation**
- C) Jaundice
- D) Seizures

 **Hint:** The doctor highlighted 'Tinnitus' (ringing in the ears) as the hallmark early warning sign of salicylate toxicity (Salicylism).

36. Aspirin prolongs bleeding time due to irreversible inhibition of:

- A) Fibrinogen
- B) Platelet Aggregation (TXA2 suppression)**
- C) Vitamin K
- D) Clotting Factors

 **Hint:** Aspirin covalently binds COX-1 in platelets, permanently disabling their ability to produce Thromboxane A2, a critical aggregator.

37. Patients scheduled for surgery should stop Aspirin how many days prior?

- A) 1 day
- B) 3 days
- C) 7 days**
- D) 14 days



Hint: Because the platelet inhibition is irreversible, it takes about 7 days (the lifespan of a platelet) for the body to generate enough new, functional platelets.

38. Aspirin can displace other highly protein-bound drugs from plasma albumin. Which drug interaction increases bleeding risk?

- A) Aspirin + Paracetamol
- B) Aspirin + Warfarin**
- C) Aspirin + Ibuprofen
- D) Aspirin + Amoxicillin



Hint: Aspirin heavily displaces Warfarin from plasma proteins. The sudden increase in free Warfarin combined with Aspirin's antiplatelet effect causes severe bleeding.

39. Management of severe Aspirin overdose involves alkalinization of urine using:

- A) Ammonium Chloride
- B) Sodium Bicarbonate**
- C) Acetic Acid
- D) Normal Saline



Hint: Aspirin is a weak acid. Alkalinizing the urine with IV Sodium Bicarbonate converts it to its ionized form, which cannot cross membranes and is rapidly excreted.

40. Why does alkalinization of urine enhance Aspirin excretion?

- A) It increases Aspirin reabsorption
- B) Ionized Aspirin cannot be reabsorbed and is trapped in urine**
- C) It increases glomerular filtration rate
- D) It converts Aspirin to Salicylic acid

 **Hint:** This principle is called 'Ion Trapping'. The charged (ionized) drug is water-soluble and cannot diffuse back through the lipid renal tubules into the blood.

41. The toxic metabolite of Paracetamol produced by CYP450 is:

- A) Glucuronide
- B) Sulfate
- C) NAPQI**
- D) Acetate

 **Hint:** A small fraction of Paracetamol is metabolized by CYP450 into the highly reactive and dangerous intermediate NAPQI (N-acetyl-p-benzoquinone imine).

42. Under normal conditions, NAPQI is neutralized by:

- A) Albumin
- B) Glutathione**
- C) Cytochrome P450
- D) Bilirubin

 **Hint:** The liver uses Glutathione, which contains sulfhydryl groups, to bind to and safely neutralize NAPQI before it can damage cells.

43. In Paracetamol overdose, liver failure occurs because:

A) Conjugation pathways are saturated and Glutathione is depleted

- B) CYP450 is inhibited
- C) Renal excretion fails
- D) Biliary obstruction occurs

 **Hint:** In massive overdose, standard pathways (glucuronidation/sulfation) saturate, pushing more drug to CYP450. NAPQI spikes, depletes all Glutathione, and then destroys hepatocytes.

44. The specific antidote for Paracetamol overdose is:

A) Atropine

B) N-acetylcysteine (NAC)

- C) Flumazenil
- D) Protamine Sulfate

 **Hint:** N-acetylcysteine (NAC) acts as a sulfhydryl group donor, essentially acting as a substitute for Glutathione to detoxify NAPQI.

45. N-acetylcysteine works by:

A) Inhibiting CYP450

B) Replenishing Glutathione and binding NAPQI

- C) Increasing renal clearance
- D) Blocking opioid receptors

 **Hint:** It restores the liver's depleted Glutathione stores, allowing the safe clearance of the toxic NAPQI metabolite.

46. For maximum efficacy, NAC must be administered within how many hours of ingestion?

A) 2-4 hours

B) 8-10 hours

C) 24 hours

D) 48 hours



Hint: To prevent irreversible hepatic necrosis, N-acetylcysteine should ideally be given within 8-10 hours of the paracetamol overdose.

47. Gout is caused by crystallization of which substance in joints?

A) Calcium Pyrophosphate

B) Sodium Urate

C) Cholesterol

D) Oxalate



Hint: Hyperuricemia leads to the deposition of Monosodium Urate crystals, particularly in the big toe, triggering a severe inflammatory response.

48. The inflammatory response in acute gout is triggered when neutrophils engulf crystals and rupture their:

A) Mitochondria

B) Lysosomes

C) Nucleus

D) Ribosomes



Hint: Neutrophils phagocytize the sharp urate crystals, which pierce their lysosomes. The released hydrolytic enzymes destroy tissue and cause massive inflammation.

49. First-line drugs for an Acute Gout Attack include:

A) Allopurinol and Probenecid

B) Indomethacin, Naproxen, or Colchicine

C) Aspirin and Paracetamol

D) Methotrexate



Hint: Acute gout requires immediate anti-inflammatory action. NSAIDs (Indomethacin) and Colchicine are the standard. Allopurinol is strictly for chronic maintenance.

50. Why is Aspirin contraindicated in Acute Gout attacks?

A) It causes GI bleeding

B) Low doses reduce uric acid excretion, worsening Gout

C) It interacts with Colchicine

D) It causes renal failure



Hint: Aspirin, especially at lower doses, competes with uric acid for the organic acid secretory transporters in the kidney, thereby increasing blood uric acid levels.